Task Description

Loads Processing for Static Strength Analysis — Turbine Rear Structure

This document is intended to be read by the engineer and sets the task scope.

1. Background

You are a stress engineer working on the static strength analysis of a **Turbine Rear Structure (TRS)** in an aero engine. The TRS is a structural component after the Low Pressure Turbine that transfers loads among its neighbouring components: LPT Case, Bearing House, Nozzle, Plug and three lugs (port, starboard and failsafe — active only when the port or starboard lugs fail).

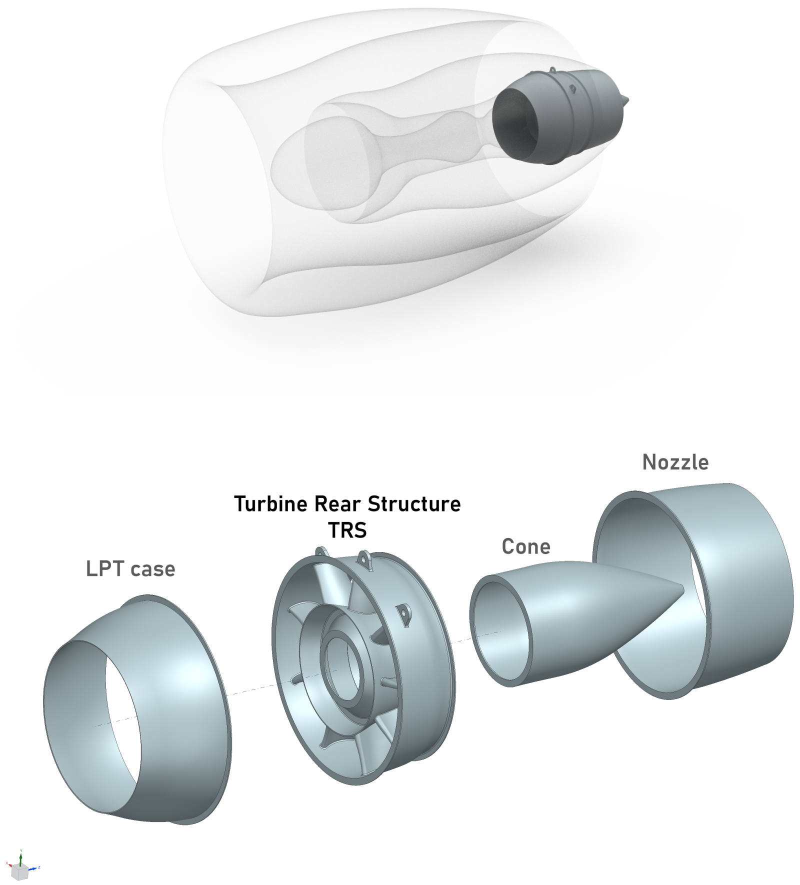

Figure 1: (top) TRS and neighbouring components positioning on the jet engine and (bottom) neighbouring components in CAD exploded view

The programme is in the **certification phase**. The OEM customer has delivered a new loads update. Your task is to process these loads into input files that can be applied to the Finite Element Model (FEM) for structural analysis. The FE mesh and solver scripts have already been set up by another team member — running the FE model is not part of this task. The mesh geometry and interface point locations are shown in Figure 2. It also shows the equilibrium example of a load case when forces are applied at the interface points.

There is no change in the material or geometrical details of the components.

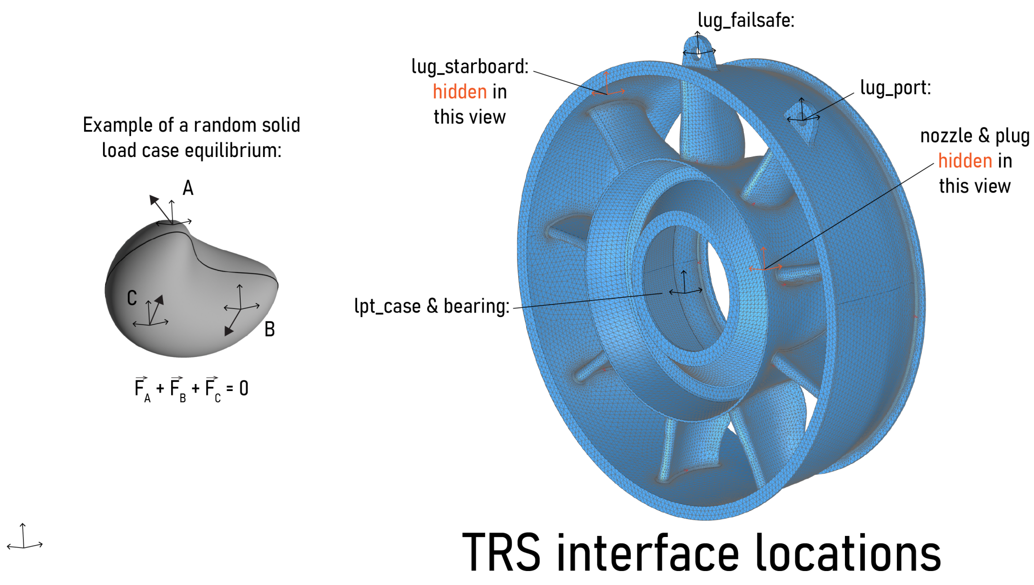

Figure 2: TRS interface locations showing one loadcase in equilibrium

2. Detailed Task Description

Your task is to perform part of the structural strength analysis, including the loads processing, ansys scirpt run and the submission of the FE model to the HPC cluster.

Make use of the certified tool `ductile-loads`, available as a Python package on PyPI (use uv). Notice that the customer has delivered the files in a different format (YAML) than expected by the tool (JSON). Part of your task is to bypass this limitation.

OEM Correction Notice

The OEM has informed that due to a human error in the Whole Engine Model export, all `Fx` force components in this delivery are understated. Apply a **1.04 multiplication factor to all Fx forces** at every interface point before processing. The OEM has chosen not to issue a revised delivery; the correction shall be applied by the component owner during processing.

3. Available Resources

The following resources are available:

- **This document** — a printed copy for you and also available in the working directory.
- **Strength Analysis Design Practive 1000** — Overall product document describing all the steps, your task is a subset of all. It is important because it points to all other subdocuments tools available to be used, beyond this document.
- `ductile-loads` — the certified processing tool, available as a Python package on PyPI. The agent can access its documentation online. The documentation is also open in your browser for your reference.
- **Previous analysis** — scripts, inputs and outputs in the `previous_run/` folder for reference.

Additionally, you or the agent can browse the internet and install Python packages via `uv pip install` *`package`*.

At any time during the task execution you can ask the researcher for help or clarification, *even half way during the execution of the task.*

4. Before You Start

Take your time

This briefing time does not count towards the task duration. Use it to familiarise yourself with the task, the design practice, the tool documentation, and the available files. Check that Claude Code is running in the VS Code terminal. Feel free to explore the working directory, browse the `ductile-loads` documentation in your browser, and review the previous analysis. Ask the researcher any questions about the task, the methodology, or the tools — there are no wrong questions. We will start the timed activity only when you confirm that you are ready.